

**Faculty of Computing & Information Technology
Department of Computer Science**



FINAL YEAR PROJECT PROPOSAL

TOPIC:

**Smart Network Diagnostic Mobile Application for ISP and Local
Network Issue Detection**

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Project Title

Mobile App for Network Speed Diagnosis

1. What This Project Does (In Simple Words)

This project is a mobile application that not only checks internet speed but also finds the real reason for slow or unstable internet.

Instead of just showing numbers, the app diagnoses the problem and tells the user whether the issue is caused by:

- Internet Service Provider (ISP)
- Wi-Fi Router
- User's Mobile Device

It also provides solutions and suggestions to fix the issue.

2. Why This Project Is Needed

Most users face slow internet but:

- Don't know who is responsible
- Restart the router unnecessarily
- Call ISP without reason

Existing apps show speed only and do not explain the cause.

This project solves that problem by identifying the root cause.

3. How the System Works (Complete Flow)

Step 1: User Starts Diagnosis

- User opens the app and clicks "Start Network Diagnosis"

Step 2: Speed & Latency Testing

The app measures:

- Download speed
- Upload speed
- Ping (latency)

Using test servers and HTTP requests.

Step 3: Network Quality Analysis

The app checks:

- Packet loss
- Jitter
- Connection stability
- Wi-Fi signal strength (RSSI)

Step 4: Network Comparison

The app compares:

- Wi-Fi vs Mobile Data
- Near router vs far router
- Multiple test servers
- Different time intervals

Step 5: Device Performance Check

The app analyzes:

- CPU usage
- RAM usage
- Background apps using internet
- Battery temperature

Step 6: Diagnosis Decision Engine

Using rule-based logic or AI, the app decides:

Condition	Diagnosis
Wi-Fi slow, Mobile fast	Router issue
Both networks slow	ISP issue
Network OK, device overloaded	Device issue

Step 7: Result & Recommendation

The app shows:

- Problem detected
- Cause identified
- Suggested solution

Example:

“Slow speed caused by weak Wi-Fi signal. Move closer to router or restart it.”

4. System Architecture (Logical View)

Mobile App:

- UI (User Interface)
- Network testing module
- Device monitoring module

Decision Engine:

- Rule-based logic
- Threshold analysis

Result Module:

- Diagnosis
- Suggestions

Optional Backend/Cloud:

- Store test history
 - Improve analysis over time
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5. Technologies Used

Frontend (Mobile App):

- Flutter (Dart) recommended
- Android (Java / Kotlin)
- React Native

Networking & Testing:

- Speed test APIs
- Ping & latency checks
- HTTP requests
- Traceroute

Device Monitoring:

- Android system APIs
- Battery & CPU monitoring
- Network usage stats

Backend (Optional):

- Firebase
- Node.js
- REST APIs

Used for user history, analytics, and reports.

AI / Logic Layer:

- Rule-based system (if–else)
 - Optional ML model (Python / TensorFlow Lite)
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6. Development Features

Core Features:

- Internet speed testing
- Wi-Fi vs mobile data comparison
- Packet loss & latency measurement
- Device performance monitoring
- Root-cause diagnosis logic
- User-friendly result screen

Advanced Features (Optional):

- Test history & reports
- Graphical analysis (charts)
- AI-based prediction
- Auto suggestions
- Network quality score

UI Features:

- Start/Stop diagnosis button
 - Real-time progress display
 - Clear messages
 - Simple and clean layout
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7. Development Modules Breakdown

Module	Description
UI Module	Screens, buttons, charts
Network Test Module	Speed, ping, packet loss
Device Monitor	CPU, RAM, background apps
Decision Engine	Rule-based logic
Recommendation Module	Solutions
Data Storage	Test history

8. Security & Privacy Considerations

- No personal data stored
 - Anonymous testing
 - Secure API communication
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9. Future Enhancements

- AI learning model
 - ISP quality ranking
 - Router auto-optimization
 - 5G performance analysis
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Conclusion

This project is a smart mobile application that diagnoses internet performance issues by analyzing network, router, and device parameters. It provides users with actionable solutions to improve their internet experience, making network troubleshooting simple and efficient.
